Lingkang Jin, Ph.D.

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RESEARCH PROFILE

Energy-systems researcher working at the interface of **mathematical optimization, multi-physics asset modelling and machine learning** for power and multi-energy systems. My work builds decision-support tools for grid operators and policymakers, in particular for **grid congestion management**, energy storage (Li-ion batteries, hydrogen) and **AI surrogates** that make security-constrained optimization tractable. I combine academic output with real-world deployment in Dutch demonstrators together with distribution system operators (Enexis, Stedin) and industry (DNV, Phase to Phase, Technolution).

13	**13**	**12**	**10**	**3**	**5**
journal articles (8 first-authored)	conference papers	h-index (Google Scholar)	MSc theses supervised	PhD students co-supervised	university courses teaching involved

Task leader in 2 RVO-MOOI and 1 Erasmus+ project · research in Italy, Denmark and the Netherlands

ACADEMIC POSITIONS

01/2024 – present **Postdoctoral Researcher**
Electrical Energy Systems (EES) group, Department of Electrical Engineering, Eindhoven University of Technology, Eindhoven (NL)
- Task leader in the RVO-MOOI projects **ORKEST** (with Stedin) and **NO-GIZMOS** (with Enexis) on grid congestion management; leader of the optimization module (Result 4) of ORKEST.
- Co-supervision of 3 PhD students and supervision of 5 MSc theses; co-lecturer of *Electricity Markets: Modeling and Optimization*.
- Task leader (Tasks 2.3 and 4.2) in the Erasmus+ project **SG-SKILL**; member of the EES group's strategic funding task force; proposal writing.

03/2023 – 07/2023 **Visiting PhD Researcher** (Erasmus+ Campus World grant)
Electrical Energy Systems group, Eindhoven University of Technology, Eindhoven (NL)
Host: Dr. Christina Papadimitriou. Output: battery degradation as a KPI for multi-objective optimization [J9].

03/2022 – 09/2022 **Visiting PhD Researcher** (Erasmus+ Campus World grant)
DTU Energy, Technical University of Denmark, Copenhagen (DK)
Host: Prof. Henrik Lund Frandsen; co-host: Dr. Rafael Nogueira Nakashima. Output: thermo-electrochemical modelling of alkaline electrolysis [J11, C3].

EDUCATION

10/2020 – 03/2024 **Ph.D. in Energy Systems**
Department of Industrial Engineering and Mathematical Sciences, Università Politecnica delle Marche, Ancona (IT)
Thesis: *Energy storage in multi-energy carrier communities: Li-ion batteries and hydrogen multi-physical details for integration into the planning stage*
Supervisor: Prof. Gabriele Comodi. **Finalist, AIMSEA best PhD thesis award 2024.**

09/2017 – 10/2019 **M.Sc. in Mechanical Engineering**, grade **110/110 cum laude**
Università Politecnica delle Marche, Ancona (IT)
Thesis: *Feasibility study and preliminary design of an offshore wind farm in the Mediterranean Sea: Wind resource assessment*
Supervisor: Prof. Renato Ricci

09/2014 – 07/2017 **B.Sc. in Mechanical Engineering**, grade **106/110**
Università Politecnica delle Marche, Ancona (IT)
Thesis: *Sensitivity analysis of a blind processing method for continuous scanning laser Doppler vibrometry*
Supervisor: Prof. Paolo Castellini; co-supervisors: Prof. Milena Martarelli, Prof. Paolo Chiariotti

RESEARCH PROJECTS AND FUNDING

2026 – present **SG-SKILL** – Skill-based education for smart-grid job profiles. *Erasmus+, European Union.*
Role: leader of Tasks 2.3 and 4.2; contributor to WP2 and WP3. 15 partners incl. E.DSO, Iberdrola, T&D Europe, RWTH Aachen, Politecnico di Milano.

2025 – present **ORKEST** – Optimal integration of network flexibility and asset intelligence to increase large-scale integration of RES while maintaining reliability. *RVO-MOOI (Netherlands Enterprise Agency).*
Role: leader of Result 4, development and integration of the optimization module. Partners: Stedin, DNV, Netbeheer Nederland, Phase to Phase, Technolution.

2024 – 2025 **NO-GIZMOS** – Grid optimization for large-scale integration of solar and wind power through storage and software. *RVO-MOOI.*
Role: research, development and deployment of the day-ahead home-battery steering algorithm (5–6% grid relief in a low-voltage demonstrator) [C13, C12]. Partners: Enexis, CGI, energieKansen, Hanze UAS.

2023 – 2024 **GRETHA** – A novel GReen Energy Technology based on fuel cells, Hydrogen And renewables. *Italian Ministry project.*

2020 – 2023 **eNeuron** – Optimising the design and operation of local energy communities based on multi-carrier energy systems. *Horizon 2020.*
Role: research support for WP2 (18 partners incl. EPRI, SINTEF, TECNALIA, EDP Labelec, ENEA) [J5, J6].

2020 – 2022 **Ricerca di Sistema Elettrico** – Methodology for the optimal energy-economic planning of a multi-energy system energy community in the medium to long term. *Italian Ministry project* [J7].

2022 **RECITY** – Resilient City: Everyday Revolution. *Italian Ministry project.*

AWARDS AND GRANTS

2026 **Eurotech Visiting Researcher Programme** grant for a research visit to DTU Wind and Energy Systems (DK)

2026 **Finalist, Best Paper Award**, TU/e Postdoc Association (400+ postdocs)

2024 **Finalist, best PhD thesis award**, AIMSEA (Italian Association of Turbomachinery and Energy Systems)

2023 **Erasmus+ Campus World** grant for a research visit to the EES group, TU/e (NL)

2022 **Erasmus+ Campus World** grant for a research visit to DTU Energy (DK)

2014 *In Studiis Laus*, Rotary Club Senigallia, and *Premio Due Valli*, Associazione GIO Misa e Nevola, for excellence in the high-school final exam

TEACHING

Years	Course (programme)	Role	Institution
2025 – 2026	*Electricity Markets: Modeling and Optimization* (5LEP0, MSc EE)	Co-lecturer, assessor	TU/e
2025 – 2026	*System Integration Project* (5LEF0, MSc SET)	Lecturer, master class	TU/e
2024 – 2026	*Electrical Power Systems for EE* (5EWD0, MSc EE)	Guest lecturer	TU/e
2022 – 2024	*Energy Storages and their Management* (MSc ME)	Guest lecturer, master class	UNIVPM
2023	*Co-generation and White Certificates* (MSc ME)	Guest lecturer	UNIVPM

- **Electricity Markets (5LEP0):** lectures and hands-on Jupyter sessions on day-ahead market clearing, single/dual imbalance pricing and renewable producers' market participation under uncertainty (CVaR stochastic programming). Course evaluation 2025/26: excellent.
- **System Integration Project (5LEF0):** master class on how to build optimization models: mathematical formulation, modular model architecture and computational exercises.
- **Electrical Power Systems for EE (5EWD0):** lectures on energy storage technologies, their technical specifications and their role in future power systems.
- **UNIVPM:** master classes on Li-ion, hydrogen and supercapacitor storage and on Python/Pyomo optimization models; lecture on co-generation plants.
- **Teaching qualification:** University Teaching Qualification (UTQ/BKO) in progress at TU/e. Planned continued involvement in 5LEP0, 5LEF0 and 5EWD0.

SUPERVISION

MSc theses (10, of whom 4 continued to a PhD)

→ PhD continued to a PhD ↳ co-supervision with industry or another research group

[S10] **Shuai Feng** → PhD, *Grid-code-compliant hybrid renewable energy solutions for data centers: Optimal sizing under post-fault active power requirements*, TU/e, 2026.
↳ In collaboration with **Shell**, company supervisor Dr. Yin Sun.

[S9] **Yuanchun Chen**, *Data-driven efficiency parametrization from operational telemetry: Improved state-of-energy tracking and aging-aware dispatch for battery energy storage systems*, MSc Sustainable Energy Technology, TU/e, 2026.
↳ In collaboration with **Zympler**.

[S8] **Chengyuan Guan**, *Optimal sizing of hybrid renewable energy solutions for data centers: Case studies of non-firm grid, reduced- and off-grid scenarios*, MSc Sustainable Energy Technology, TU/e, 2025.
↳ In collaboration with **Shell**, company supervisor Dr. Yin Sun.

[S7] **Dennis Hollanders**, *Graph neural networks for distribution network reconfiguration optimization*, Information Systems group, Dept. of Industrial Engineering & Innovation Sciences, TU/e, 2025.
↳ Co-supervised with Dr. Laurens Bliek (Information Systems group, TU/e).

[S6] **Stefan De Lange** → PhD, *Hybrid heat pump optimization for flexibility provision: Modeling and simulation*, Dept. of Electrical Engineering, TU/e, 2024 (co-author of [C8, C11]).

[S5] **Fabian Caipa Cure**, *Adaptive distributionally robust optimization for residential energy management under non-firm capacity contracts*, MSc Sustainable Energy Technology, TU/e, 2024.

[S4] **Francesco Panara**, *Study of hydrogen-to-power systems: State of the art of alkaline and PEM fuel cells and performance evaluation through Python modelling*, UNIVPM, 2023.

[S3] **Filippo Onori** → PhD, *Design and management of a battery energy storage system (BESS) to provide flexibility services to the national electricity grid*, UNIVPM, 2023.

[S2] **Francesca Mennilli** → PhD, *Study of power-to-hydrogen systems: State of the art of the electrolyser and its modelling using Python*, UNIVPM, 2022 (co-author of [J8, J10, C1, C7]).

[S1] **Luca Ciotti**, *Study of power-to-hydrogen systems: State of the art of the storage and its modelling using Python*, UNIVPM, 2022.

PhD co-supervision (3)

[P3] **Ranier Alexsander Arruda Moura**, *Copula-based synthetic profile generation for MV distribution networks*, TU/e, 2025 – present.

[P2] **Niek Brekelmans**, *MV/LV power transformer reliability and aging analysis for system operation*, TU/e, 2025 – present.

[P1] **Yifan Zhang**, *MV/LV cable reliability and aging analysis for system operation*, TU/e and DNV, 2025 – present.

PUBLICATIONS

13 journal articles (8 first-authored; 10 in Q1 journals), 13 conference papers (5 first-authored), 1 preprint; h-index 12 (Google Scholar).

Journal articles

Q1 Q2 journal quartile (Scimago SJR, main subject category)

[J13] **L. Jin**, L. Bliek, and N. G. Paterakis, "Learning Active Constraints for Bilevel Optimization: Evaluating Classification Metrics on Real-World Distribution System Data," *IEEE Transactions on Smart Grid* Q1, vol. 17, no. 2, pp. 1303–1314, 2026. doi:10.1109/TSG.2025.3632419

[J12] **L. Jin**, L. Bliek, and N. G. Paterakis, "Surrogate Models in Power System Scheduling Problems: Learning Active Constraints via Graph Neural Networks," *IEEE Transactions on Smart Grid* Q1, early access, 2026. doi:10.1109/TSG.2026.3681028

[J11] **L. Jin**, R. Nogueira Nakashima, G. Comodi, and H. L. Frandsen, "Alkaline electrolysis for green hydrogen production: A novel, simple model for thermo-electrochemical coupled system analysis," *Applied Thermal Engineering* Q1, vol. 262, Art. no. 125154, 2025. doi:10.1016/j.applthermaleng.2024.125154

[J10] **L. Jin**, M. Rossi, A. Monforti Ferrario, F. Mennilli, and G. Comodi, "Designing hybrid energy storage systems for steady green hydrogen production in residential areas: A GIS-based framework," *Applied Energy* Q1, vol. 389, Art. no. 125765, 2025. doi:10.1016/j.apenergy.2025.125765

[J9] **L. Jin**, M. Kazemi, G. Comodi, and C. Papadimitriou, "Assessing battery degradation as a key performance indicator for multi-objective optimization of multi-carrier energy systems," *Applied Energy* Q1, vol. 361, Art. no. 122925, 2024. doi:10.1016/j.apenergy.2024.122925

[J8] F. Mennilli, **L. Jin**, M. Rossi, and G. Comodi, "Assessment of a NaOH-based alkaline electrolyser's performance: System modelling and operating parameters optimisation," *International Journal of Hydrogen Energy* Q1, vol. 85, pp. 625–634, 2024. doi:10.1016/j.ijhydene.2024.08.175

[J7] A. Pizzuti, **L. Jin**, M. Rossi, F. Marinelli, and G. Comodi, "A novel approach for multi-stage investment decisions and dynamic variations in medium-term energy planning for multi-energy carriers community," *Applied Energy* Q1, vol. 353, Art. no. 122177, 2024. doi:10.1016/j.apenergy.2023.122177

[J6] M. Rossi, **L. Jin**, A. Monforti Ferrario, M. Di Somma, A. Buonanno, C. Papadimitriou, A. Morch, G. Graditi, and G. Comodi, "Energy Hub and Micro-Energy Hub Architecture in Integrated Local Energy Communities: Enabling Technologies and Energy Planning Tools," *Energies* Q2, vol. 17, no. 19, Art. no. 4813, 2024. doi:10.3390/en17194813

[J5] **L. Jin**, M. Rossi, L. Ciabattoni, M. Di Somma, G. Graditi, and G. Comodi, "Environmental constrained medium-term energy planning: The case study of an Italian university campus as a multi-carrier local energy community," *Energy Conversion and Management* Q1, vol. 278, Art. no. 116701, 2023. doi:10.1016/j.enconman.2023.116701

[J4] **L. Jin**, M. Rossi, A. Monforti Ferrario, J. C. Alberizzi, M. Renzi, and G. Comodi, "Integration of battery and hydrogen energy storage systems with small-scale hydropower plants in off-grid local energy communities," *Energy Conversion and Management* Q1, vol. 286, Art. no. 117019, 2023. doi:10.1016/j.enconman.2023.117019

[J3] E. Marchegiani, F. Ferracuti, A. Monteriù, **L. Jin**, M. Rossi, G. Comodi, and L. Ciabattoni, "Li-ion battery aging model robustness: An analysis using univariate and multivariate techniques," *Journal of Energy Storage* Q1, vol. 72, Art. no. 108591, 2023. doi:10.1016/j.est.2023.108591

[J2] **L. Jin**, A. Monforti Ferrario, V. Cigolotti, and G. Comodi, "Evaluation of the impact of green hydrogen blending scenarios in the Italian gas network: Optimal design and dynamic simulation of operation

strategies," *Renewable and Sustainable Energy Transition*, vol. 2, Art. no. 100022, 2022. doi:10.1016/j.rset.2022.100022

[J1] A. Nadeem, M. Rossi, E. Corradi, **L. Jin**, G. Comodi, and N. A. Sheikh, "Energy-Environmental Planning of Electric Vehicles (EVs): A Case Study of the National Energy System of Pakistan," *Energies* Q2, vol. 15, no. 9, Art. no. 3054, 2022. doi:10.3390/en15093054

Conference papers

[C13] **L. Jin**, S. Zhan, S. Cudjoe, and N. G. Paterakis, "Empowering Low-Voltage Grids: Real-World Implementation of Home Batteries for Effective Congestion Management," in *Proc. 2025 IEEE Kiel PowerTech*, pp. 1–6, 2025. doi:10.1109/PowerTech59965.2025.11180463

[C12] **L. Jin**, S. Zhan, H. Zhang, and N. G. Paterakis, "Multi-Objective Congestion Management in Low-Voltage Grids: Coordinating End-Users and System Operators for Real-World Implementation," in *Proc. 2025 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT Europe)*, pp. 1–5, 2025. doi:10.1109/ISGTEurope64741.2025.11305401

[C11] X. Li, J. J. Markus, S. De Lange, **L. Jin**, K. Kok, and N. G. Paterakis, "Hybrid Heat Pump Flexibility Allocation: Quantified Thermal Comfort-Based Congestion Management," in *Proc. 2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, pp. 1–6, 2025. doi:10.1109/EEEIC/ICPSEurope64998.2025.11169153

[C10] H. Zhang, **L. Jin**, K. Kok, and N. G. Paterakis, "Surrogate Model-Based Reinforcement Learning for Bidding Strategies in Local Flexibility Markets," in *Proc. 2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, pp. 1–6, 2025. doi:10.1109/EEEIC/ICPSEurope64998.2025.11169071

[C9] M. Di Somma, **L. Jin**, N. Bianco, and C. Papadimitriou, "Advancing Multi-Energy Hub Design: an Integrated Approach for Optimizing Residential Clusters in High RES Penetration Scenarios," in *Proc. 2024 International Conference on Smart Energy Systems and Technologies (SEST)*, pp. 1–6, 2024. doi:10.1109/SEST61601.2024.10694519

[C8] **L. Jin**, X. Li, S. De Lange, H. Slootweg, and N. G. Paterakis, "Response Allocation of Domestic Hybrid Heat Pumps Flexibility for Congestion Management," in *Proc. 2024 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*, pp. 1–5, 2024. doi:10.1109/ISGTEUROPE62998.2024.10863630

[C7] F. Mennilli, **L. Jin**, M. Rossi, and G. Comodi, "A Fitting Process for the Optimal Modelling of an Anion Exchange Membrane (AEM) Electrolyser," in *Proc. ASME Turbo Expo 2024: Power for Land, Sea, and Air*, vol. 2, Art. no. V002T03A017, 2024.

[C6] J. C. Alberizzi, M. A. Pérez Estevez, M. Renzi, **L. Jin**, M. Rossi, and A. Alberizzi, "Optimal Management of a Hydro – Wind Energy System with Hydrogen Storage," in *Proc. 2023 12th International Conference on Power Science and Engineering (ICPSE)*, pp. 46–50, 2023. doi:10.1109/ICPSE59506.2023.10329307

[C5] A. Bechmann, T. Barlas, H. L. Frandsen, **L. Jin**, and R. N. Nakashima, "The Hydrogen Wind Turbine: Design of a wind turbine optimised for hydrogen production," *Journal of Physics: Conference Series*, vol. 2507, no. 1, Art. no. 012010, 2023. doi:10.1088/1742-6596/2507/1/012010

[C4] M. Di Somma, A. Buonanno, M. Caliano, **L. Jin**, M. Rossi, G. Graditi, and G. Comodi, "Stochastic energy management for the Italian UNIVPM campus as a multi-carrier energy hub participating in the day-ahead market," in *Proc. IEEE EUROCON 2023 - 20th International Conference on Smart Technologies*, pp. 251–256, 2023. doi:10.1109/EUROCON56442.2023.10199000

[C3] **L. Jin**, R. N. Nakashima, H. L. Frandsen, and G. Comodi, "Alkaline Electrolysis for Green Hydrogen Production: Techno-Economic Analysis of Temperature Influence and Control," in *Proc. 36th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS)*, 2023. doi:10.52202/069564-0082

[C2] **L. Jin**, M. Rossi, F. Caresana, L. Pelagalli, and G. Comodi, "Metal hydrides in hydrogen storage: optimization of dynamic control strategies," *Journal of Physics: Conference Series*, vol. 2648, no. 1, Art. no. 012056, 2023. doi:10.1088/1742-6596/2648/1/012056

[C1] F. Mennilli, **L. Jin**, M. Rossi, A. Mugnini, and G. Comodi, "Energy analysis of a hydrogen integrated system in the residential sector," *Journal of Physics: Conference Series*, vol. 2648, no. 1, Art. no. 012057, 2023. doi:10.1088/1742-6596/2648/1/012057

Preprints

[R1] S. Zhan, **L. Jin**, H. Zhang, and N. G. Paterakis, "Real-time Measurement-based Optimization for Distribution System Operation Considering Battery Voltage and Thermal Constraints," *arXiv preprint*, arXiv:2510.16408, 2025. https://arxiv.org/abs/2510.16408

ACADEMIC SERVICE AND OUTREACH

Organization and leadership

- Co-organizer and chair of the workshop *Congestion Management in the Netherlands 2025*, TU/e.
- EIRES (Eindhoven Institute for Renewable Energy Systems) representative at the EU Sustainable Energy Week (EUSEW) Energy Fair 2026.
- Member of the strategic funding task force of the EES group, TU/e.

Reviewing

- **Journals:** IEEE Transactions on Smart Grid, IEEE Transactions on Intelligent Transportation Systems, IEEE Systems Journal, Applied Energy, Energy Conversion and Management, Journal of Energy Storage, Sustainable Energy, Grids and Networks, Sustainability.
- **Conferences:** IEEE ISGT Europe, IEEE PowerTech, SEST, ECOS.

Invited talks and dissemination

- Speaker at 20+ venues for academic, industrial and general audiences, incl. Enlit Europe, EUSEW and DTU Wind and Energy Systems.

Memberships

- IEEE Power & Energy Society (since 2024); Italian Association of Turbomachinery and Energy Systems, AIMSEA (2020 – 2024); Order of Engineers, Italy (2020 – 2024); SAE International (2018 – 2019).

LANGUAGES AND CERTIFICATES

Languages	Chinese (native), Italian (native), English (C1, professional working language), Dutch (A2, learning)
2022	**Cambridge C1 Advanced (CAE)**, English
2019	**Cambridge B2 First (FCE)**, English
2019	**Professional engineering licence** (Abilitazione all'esercizio della professione di Ingegnere, Sezione A), Italy

RESEARCH STATEMENT

Vision

Energy-system models are the everyday decision tools of grid operators, market operators and policymakers: they clear markets, schedule assets and coordinate network operators with end users. These models are now under pressure. Flexible assets such as batteries, electrolysers and heat pumps are being connected faster than grids can be reinforced, and inaccurate models translate directly into missed connections, long waiting lists for grid access and avoidable social costs.

To remain tractable, current practice relies mainly on linear and mixed-integer linear programming and describes assets from a purely electrical viewpoint, in which power and energy are the only state variables. The thermal and electrochemical dynamics that determine efficiency, degradation and safe operating limits are largely neglected.

My research vision is to close this gap with a **bottom-up approach**: embedding the essential multi-physics behaviour of energy assets into system-level planning and operation problems, and using **machine learning to keep these richer models fast enough** for real-world decisions. The result is a hybrid modelling framework – physics-based where accuracy matters, AI-accelerated where computation is the bottleneck – that delivers reliable decision-support tools for grid operators and policymakers.

Past research

My research so far follows three interconnected directions, all addressing the question of how complex energy assets can be integrated into system-level decisions.

1. **System-level optimization.** I developed deterministic, bilevel and multi-objective optimization models (LP, MILP, QP) for multi-energy systems, quantifying the synergies between energy carriers in local energy communities [J4, J5, J7, C9]. At TU/e I moved to distribution-grid congestion management from the perspective of both network operators and end users, with validation in real Dutch demonstrators [C8, C13, C12].
2. **Multi-physics asset modelling.** Supported by competitive mobility grants, I characterized the thermal and electrochemical behaviour of alkaline electrolysers at DTU (2022) [J11, C3], and quantified Li-ion degradation as an implicit operating cost at TU/e (2023) [J9]. My PhD thesis integrates these results and their system-level impact.
3. **Machine learning for tractability.** Detailed asset physics introduces non-linearities and bilevel structures that are computationally expensive. I therefore developed ML surrogates that learn the active constraints of power-system problems from real distribution-grid data [J13], Graph Neural Network surrogates for scheduling problems [J12] and surrogate-based reinforcement learning for bidding in local flexibility markets [C10], and I contributed to measurement-based real-time operation that respects battery voltage and thermal limits [R1].

Future research plan

I will build on this momentum along three lines that converge into one framework:

- **Asset physics in system operation.** Extend multi-physics modelling to grid assets, starting with the aging of power transformers and cables in the daily operation of distribution system operators (PhD projects P1 and P2, with DNV and Stedin).
- **Generalizable AI surrogates.** Graph Neural Network surrogates have proven effective for deterministic problems [J12]. I will investigate transfer learning across network topologies and the extension of surrogates to stochastic and robust formulations.
- **Scenario generation and uncertainty quantification.** The quality of input data drives both real-world alignment and surrogate training. I will develop data-driven and generative scenario-generation methods, such as copula-based synthetic profiles (PhD project P3), tailored to each application. Energy informatics plays a central role here.

Funding plan

- **Individual grants:** NWO Talent Programme **Veni** and, at European level, the **ERC Starting Grant**.

- **Consortium grants: Horizon Europe Cluster 5** (climate, energy and mobility) and the **Clean Energy Transition Partnership**, building on my network in the Netherlands, Denmark and Italy.
- **National schemes with industry: RVO MOOI**, where I already hold task-leader roles in ORKEST and NO-GIZMOS, and NWO programmes with industrial co-funding.
- **Direct industry collaboration:** contract research with grid operators, consultancies and technology providers, with research questions driven by concrete business cases.

TEACHING AND SUPERVISION STATEMENT

Teaching vision

I see teaching as creating a **safe and positive learning environment** in which students dare to ask questions, discuss and take an active role. In my lectures I achieve this in three ways:

- **Active listening.** When a student asks a question, I first rephrase it and ask for clarification before answering. This makes sure I address the student's real difficulty, makes the question accessible to the whole group (others often share the same doubt) and gives me direct insight into how students interpret the material. Taking questions seriously shows students that their contribution is valued, which encourages further participation.
- **Guiding attention.** I point at the exact part of an equation or piece of code I am explaining and move through the classroom. This helps students follow complex content and reduces the distance between teacher and students, so it is easier for them to speak up.
- **From concept to implementation.** I move step by step from slides for the concepts, to written notes for the mathematical formulation, to interactive Jupyter notebooks for the implementation. Students approach the topic at increasing levels of concreteness, and the lecture connects directly to their course assignments.

Reflection is part of my practice. A recorded lecture and peer observations within my University Teaching Qualification (UTQ) showed that the interaction with students was a clear strength, and also gave me concrete points to develop:

- **Structure:** I divide the content into short blocks, each followed by a planned moment for questions and discussion, so that interaction stays valuable without students losing track of the lecture.
- **Prior knowledge:** I start each lecture with a short recap of earlier material linked to the new topic, so that all students start from the same foundation.
- **Delivery:** a well-structured lesson plan, clear key points on the slides and growing experience help me deliver content calmly and confidently from the first minute.

Supervision vision

My aim is to help students become **independent researchers** who can think constructively and critically about their own work. I do this by challenging them to defend and justify their choices, while giving them the tools and direction they need to do so themselves. Supervision, in my view, means helping students take ownership of their project rather than taking it over from them. This prepares them for their future as researchers and professionals: of the ten MSc students I supervised, four continued to a PhD, and several co-authored peer-reviewed publications [J8, J10, C1, C7, C8, C11].

STATEMENT ON AI IN RESEARCH AND DEVELOPMENT

AI in research

Large language models and AI assistants (e.g. ChatGPT, Gemini, Claude) are already part of research practice. I believe their use is inevitable and should be guided by critical thinking: they support, but never replace, the researcher's own ideas, and every output must be reviewed, verified and tested by the researcher. I use AI tools only where they add value, because many tasks are essential to keep core research skills trained:

- **Idea generation:** AI may suggest directions, but their validity and feasibility must be elaborated and verified by the researcher.
- **Figures and presentations:** turning a complex problem into a clear illustration is a key communication skill that I keep in my own hands.

- **Writing:** AI may improve clarity, but only after the paper structure and first draft have been written independently.

AI development practice

1. **Problem statement and mathematical formulation:** minimal to no AI involvement, so that the theoretical foundations are original and sound.
2. **Coding and implementation:** AI support is used mainly here, always based on the validated formulation from the previous step.
3. **Validation:** a modular, step-wise validation process in which every AI-assisted step is checked against the formulation.
4. **Manuscript preparation:** draft and structure are written without AI; AI is used for final proofreading only. Figures are made without AI to guarantee their reliability. AI may suggest references, but every source is read and verified personally.